Task 4 – BOM

IR Sensor Array (QTR-8 RC)

- We can use a premade ir sensor array for this project instead of manually taking multiple ir sensor modules … basically
- It has the following advantages
    - In the premade ir array the most important advantage is that the 8 pairs come presoldered and with also equal spacing so first thing is we dont need to take the pain to measure the dinstaces and make sure all the 8 modules of ir sensors are fixed and also that there will be vey very less pain that we need to take during the calibration …
    - And also the connections will also be easy since we will get all the pins directly soldered to one singular board so the connections will be more organised and the connections will also not look that shabby
    - Link - https://www.amazon.in/Electron-QTR-8RC-Following-Reflectance-Sensor/dp/B0DYP4JP2V?mcid=8d4085bf35be3cfd955659775a985b49&tag=googleshopdes-21&linkCode=df0&hvadid=709963085510&hvpos&hvnetw=g&hvrand=11556278949578119663&hvpone&hvptwo&hvqmt&hvdev=c&hvdvcmdl&hvlocint&hvlocphy=9183877&hvtargid=pla-2422909312205&psc=1&hvocijid=11556278949578119663-B0DYP4JP2V-&hvexpln=0
    - Price – 240Rs

Microcontroller (ESP 32 WROOM 32 Devkit board)

- We can also use the chip itself .. but using the devkit board is more easy.. So devkit board it is
    - Link - https://www.amazon.in/SP-ELECTRON-ESP32-Development-ESP32-WROOM-32/dp/B0HD7QC26L?crid=2K6Z1AM6WO9VD&dib=eyJ2IjoiMSJ9.H5Qdvll8rdKi4WqCiS9VhdI3iaZbhgtQl5klI5fbAqTaHnH9PVfhgVGmASJWMuZNB3Mxu6roYG4ggagBCZ7XaCWCBcJvx_T-YgyaVu8gdqFZE-4fkuTlUJdxvgzD6RV8qxLBlogARd1CI1B5OTSbrFFtjrvCrieT-9f5GAYHtIQ6vPddJV-NWIbnqPjV12Tw1KI-4X_kkj5czKc-yjeG4fTURmEwLl-IIzfKoPTiqysDBVjEEEs0xjfvRMQ_3V4bB2NMqUA4Y4goWrmo1mJqeYysbLSQQruYxZcDzECG-SM.LJbMbiMqNHSWJKMeUsR8qRuBlAAKLQ8eyhUiBMuj1z0&dib_tag=se&keywords=esp+32+WROOM+32&qid=1792926887&s=industrial&sprefix=esp+32+wroom+3%2Cindustrial%2C385&sr=1-1
    - Price – 500Rs

Motors (2 N20 6v Metal Geared Motors{200rpm} with encoders)

- The main reason of me not using the 300 rpm motor is eventhough it is faster it is harder to tune ..and besides we are not making a line following robot where speed is the priority our priority is accuracy and easier tuning so i think the 200 rpm will do just fine
- And also compered to normal tt plasic geared motors .. the main difference is the gears (which are metal in n20) and also their operating voltages .. and also has an encoder .. to measure the roations of the motor .....
- Basically more control during turnings and intersections .. and far less pain during tuning..
- And the reason for the encoders is because we can map the maze ..
    - Link - https://www.amazon.in/200RPM-Micro-Metal-length-Encoder/dp/B0G49CC6YW?crid=3SMQO1HO4OMWI&dib=eyJ2IjoiMSJ9.IiFODmiFUn4Yi0w6gZSeuVWxcZkYJ94dEZewNoefHkrjb2bYAcOb9CeZA65lPkWxwO9VZsRXGqGKzzTBRmWnNWG_KQbTb4P9ei96Tvu59oFKAkUYg9mikC9RnTjmWncLVR-J90AWtkY4ILPZmjVmC9Qw5FfXPKLCo01mBh7klzYnrcsNNqNrUCVkkp4iEoSLWyaSXoIGLkwZiBwDuQUe47U8_HjukcBQ64aXYlXwMNqihOPqz2uqF6ul9sgEtPwVM0KUNM0Y2o3D3IFUy3bG0rhddLYJk5H5FWoKe0KPNt0.K8yGXiBtykzxvx-28_CDnresWipz3i5TDnyScKML3FM&dib_tag=se&keywords=n20+Geared+motors+200rpm&qid=1792927333&s=industrial&sprefix=n20+geared+motors+200rp%2Cindustrial%2C297&sr=1-1
    - Price 1200Rs
- For Wheels we need some wheels with more traction so something like these below...
    - Link - https://dyoraarobotics.in/product/2
    - Price 269Rs
    - Or we can get soemthing like this also - https://www.theengineerstore.in/products/n20-dc-gear-motor-wheel-42mm?variant=43720257994915&country=IN&currency=INR&utm_medium=product_sync&utm_source=google&utm_content=sag_organic&utm_campaign=sag_organic
    - Price 180Rs

Motor Driver (TB6612FNG Dual DC Motor Driver Module)

- The main reason we are using this is in line folowing bots generally .. we have rapid acceleration and suddeen breaks .. so to have better control we use the above(especially with pid controller)
- And also that the l298n is more bulky
    - Link - https://www.amazon.in/TB6612FNG-Dual-Motor-Driver-PACK/dp/B0GMVYN79G
    - Price – 400 Rs

Miscellaneous Stuff

1. LED Light
    a. For indicating that it finished the maze
    b. LED Toh club se mil hi jaayegi .. khareedne ki kya jaroorat 🥀🥀🥀

c. Link - https://www.amazon.in/Electronicspices-Ultra-Bright-White-Diode/dp/B085BMH3XZ?crid=1RMTZFUN22F45&dib=eyJ2IjoiMSJ9.-cwHRl7D3tkTZrkwgsrs1i33jua-2d8lJZRwI9BjtWLGP9ABlPwH14FOhKKCR6MuO3fQmsvU4sGltYJUJmyPzLZM9jK38sZmActQI_qFlMUIqKdrVHIHMFFXEbPPihfEoqqTIfrY_yinPvGXZcymY0wE8yc5rgmY6cWJVdBJzhoAEuNpfr8qTY4TcGxioZyBPH2rsypDF1YBz72x28HVcDoGMJ9_hbIIx8Yr7cb4CWRpoSdwUbvhTLkVSoaFMrxyupPJmOzO1FUoutb-GAmzl2HX4JfEcN9MQAjus1ui8Qk.dimrmvLh2qJJvHRXkAWKK5Me85e_ld65yIFTocHJjow&dib_tag=se&keywords=single+led+light&qid=1792956319&sprefix=single+led+ligh%2Caps%2C324&sr=8-7

d. Price – 50 Rs

2. A Switch
    a. This too .. club se hi utha lenge
    b. Link - https://www.amazon.in/SP-Illuminated-Electronics-Automotive-Appliance/dp/B0H6F78F97?crid=2I11LVZ0ZXP0V&dib=eyJ2IjoiMSJ9.7VsmIOwBhZtMOOwoROD4Bjy10WjQ5FzPBCHhzmdyUE6cMd8ftbH-x4OLg-ebYPnh6sf8isKFQFg4h1D-A_Dnizx9sy4EIqWXjzP978ocoG8jMRnQITYjNLv6H1vshoRQpmI4nYYaAY3oYaDV5O8MCxwXOIfV6Z6e9krBDPzyyOA_hjQ332DzMAGFyXUlDkkdg08v052mEobbHjn5KZoFAL_9qgDE8AInlkjpZaOfIUyjSaUCRX0Z44boyoUFcTwDDZA0nxcjtwxYmUBcHrF9fXlYpumci1YIr1l-lUO0K0Y.LFP7ZUaqg6tQw8_nfKyGr-sRr4yY7O2Nwdat6xRil6c&dib_tag=se&keywords=small+switch&qid=1792956360&sprefix=smallswithch%2Caps%2C379&sr=8-6

    c. Price – 30 Rs
3. Chasis
    a. We want it to be less bulky ..
    b. Something like below can work (My personal preference btw .. we can pick something else also ..)
    c. Link - https://www.amazon.in/Round-Acrylic-Robot-Chassis-Black/dp/B0F47K9N19?crid=6QN0RTTGZ8YM&dib=eyJ2IjoiMSJ9.856Ifh5W_4uj8KK69ib8xmKDPJgfXLcTyrZHXn6idEqs-x0SGUExU8W2lK3G4QBMDWQF16Tp7IRQs-Prsx9_vSrcQ227eSCYA87jGbkSxE0jejRvb5TOm2J7hjsgJIl_OCVjPiGDDZggFm81PFtcugPmqgUWAKtFfk8wi94JVDsOcCVWUMMnzOK3SnLUv-FnXt6uGSiwnX5Xd9u3yZGIfzMZnI7KDCx4bkdkL2mi3DSIHrHK6oWo7jJwpRaogC5Nxsw3NJnqOKoav7AyftORtuCEfm2HUui7phr1URfQtlM.4eW-1UESuptiZtQlDq627W3o_9XS9CsuDTufjrhRxdU&dib_tag=se&keywords=car+chassis&qid=1792956400&sprefix=car+chas%2Caps%2C382&sr=8-27

    d. Price – 600 Rs
4. Connector and zero pcb Board and jumper wires
    a. Link - https://www.amazon.in/UNIVERSAL-HUB-General-electronic-Experiment/dp/B0C3B8R4CX?crid=G20W320OA1I4&dib=eyJ2IjoiMSJ9.rvprrkAGoYDmTGGZcpW5-LB_SSX_qKavGwy7gbx-QGJF7kk0xkuVN1XKLsn3NYmE5XCfMSEE4FLf05Ft66lQPR0mwJN1HZngcRh31YkBpQ1sH0ZhXidhxUXohDT28-Ux4XShc341JAhvN7J5jAGsc3LcW7ViuYpdDdtCnIj8b6fFo8jx3hf4BvZofGQPE9tkjL0L3Ai7upn72W0uXyYBwrW4QPvwfK3dXA4_inPigjEg_I1Kftf6PdxaWCGeFMPDJWe9T_i

ycP7fzX8XEMrV8pmcKRtgRaxA3ttXQ4tWEWk.CCd9I4jpdmPiEeVshjx9g_u_gYI5TR9Bh1fYSFlxTaE&dib_tag=se&keywords=pcb&qid=1792956643&sprefix=pcb%2Caps%2C365&sr=8-6

b. **Link -** https://www.amazon.in/UNIVERSAL-HUB-2-54mm-Female-Connector/dp/B0DNMD4THF?crid=1J8VLXGGWHWP&dib=eyJ2IjoiMSJ9.NpiqG_GJTt7YLKhI8CJENxT9AvEhnhfPJXrrYLyOERHSGC4_kxLXNqnx0yyzk-aqy7g2cLjQEYlCnzHF__0pjGKnuop75452prkhQ4znJBnFSVYBcz9eldmtTRidzbQc4DmuNrfiCceRYMJNM6W7eL5pr_Xqyc6JkmTgwn8p1R1btbeRhk5E5Tm8h0gLLnbxJd7xVw9twV3TtPaPuX2JgTNDe5iHCTgdcy52Qe1m_cebNtl-nfkhR3N2oAA6v9CQbXuUWuf6_HwUrBcY5VvgNCYjBU1mlOp7oGow1_iabwk.z5YUZ65MlMrMsfVV9tS6T3RaRvppjT4B6pGOHPkhtww&dib_tag=se&keywords=connector+for+arduino&qid=1792956703&sprefix=connector+for+arduino%2Caps%2C336&sr=8-9

c. **Link -** https://www.amazon.in/Electronic-Spices-Single-Core-Breadboard-Jumper/dp/B0B4WPX449?crid=9X3VZ73B5WV3&dib=eyJ2IjoiMSJ9.vLBL377-brAFNacDz1hApdJzm2dtYtf3ydYQDIMqX9zDlBkcyG_U3TdCXK5vMR9_ddvE36Q8kyzVFZUWfawGi5HX2bVezUOWNIJ5wCM8SDvTGkSNGIQhMOUNQojrLdxHsjpogqVwPZ_fkp7bKSqgTcFXwIpiI6pCDEHFyLO1ZlHowV1UmZL_rZLIJaQ_HRltK4aQQLb7xKn7bq_OYg5ZD2vpYRjQvhvW7r4lnTxXE0EonLFN2yQLBu_zELQ38c8Refl21fFhmgnWo14Cw1M5T8ozV3_z9JL4XUzh_y-4SoI.2GsUc3g-NHeFeNYO9lT6TlBktI4g_z5y3Y-cKrtoV-k&dib_tag=se&keywords=wire&qid=1792956733&sprefix=wire+f%2Caps%2C351&sr=8-10

d. **Link -** https://www.amazon.in/Aptechdeals-Jumper-Wires-Female-Breadboard/dp/B074JB6SX8?crid=U5C7KWHH5QD1&dib=eyJ2IjoiMSJ9.NpiqG_GJTt7YLKhI8CJEN93BeTCKz0zXicFcYBnJD_L-pdUlzupjXnJ_2TqtdUJsGJk-8IcymmMln_8F8uswHeH1QdxGKFiKcD2USV5Tu0hDG8Fz0M_p8cqT91tMxepvU5dn19dCKcgfS0MdT-XWe8XavWfLxDYEtzJEIXGPG3RhQjtzZrpBLPuYUnEfzla6d5Q9cyMx2Wzes7IyJzkFr9NUYjIX5ypTQE4U6F_txL8FZu03rW8NE70vkxjWNH8I8ZmP3-VhhLXpIJgnOdwythShLkztNNB1MVGuvo5PlAI.9e1AUkH8El0hvo9vWiJb-emdn0oyzJONPfReXiNrTzw&dib_tag=se&keywords=jumper+wire+set&qid=1792956577&sprefix=jumper+wire+se%2Caps%2C353&sr=8-6

e. **Price – 60 + 100 + 150 + 140 = 450 Rs**

5. **Buck converters ..**
   a. **We need 2 actually one for the 5v and one for the 6v**
   b. **Link -** https://www.amazon.in/Robocraze-Lithium-Battery-Charging-Converter/dp/B07RKH7YVY?crid=1XH81LRF6O6IS&dib=eyJ2IjoiMSJ9.SdP7Hn0UC0hg9sVzZKNKRjp7VQxWtL7Ptks-qowE_IkszxkIlcpUC_XSiOTL10xQI-ea_cagnk7hvw7Dg_b255KHExF-udaxGtXKvYp0t5xakESHLX_v5TA2SA71ODqupyO4HrLPWOgsOUai85we0WdIGsTBUcaXxRRtjEBLO39qWgWPe8J6M7yf4i8BX0SOYf3TtOLdTVKKqxDoPKm8ZK-srQLm8DNyhvAv9tVzTgQ.11TuwI9FQiXiannx1MWrQQTbouhYjeY_Rq356ZCFORE&dib_tag=se&keywords=buck+converters&qid=1793003385&sprefix=buck+converter%2Caps%2C330&sr=8-3
   c. **Price – 2(240) - 480 Rs**

6. **3A Fuse ..**
    a. **Link - https://www.amazon.in/Ovicart-Glass-Fuse-Holder-3A/dp/B09X7B8BM8**
    b. Price – 180 Rs
    c. Reasoning is given where i calculated the max curent draw
7. And ye almost me bhul hi gaya(XT60 connector female and male both)
    a. Link - https://www.amazon.in/Universal-Hub-Pair-XT60-Connector/dp/B0GSBNDQ3L/525-8887240-1156925?pd_rd_w=9DD4Z&content-id=amzn1.sym.d1406b44-aa69-47e4-9270-f613e12d52dc&pf_rd_p=d1406b44-aa69-47e4-9270-f613e12d52dc&pf_rd_r=5NN851TSMB7TNE62N8GB&pd_rd_wg=lUUSA&pd_rd_r=adfcad74-263f-4b85-9296-ea022158de86&pd_rd_i=B0GSBNDQ3L&psc=1
    b. Price – 90 Rs

Total ~ 5k + 1.6k (for the battery) = 6.6k

For Battrey Refer to the Power budget and also the powerblock diagram ..